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Hydro-Thermo-Mechanical-Chemical Modelling of Venus Crust Hydration

Bachelor Thesis

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Abstract

This thesis develops a hydro-thermo-mechanical-chemical (HTMC) numerical framework to investigate potential hydration processes in the Venusian crust. Due to the lack of direct constraints on Venusian interior structure and hydration, the focus of this work is on model development rather than on deriving definitive geological conclusions. A thermodynamic database approach is implemented to efficiently compute pressure-, temperature- and hydration-dependent material properties and to control water partitioning between solid-bound and free pore fluid phases.

The analysis of the computed database reveals that most Venus lander-based compositions exhibit significant solid-phase water storage capacity, suggesting Venusian crust could act as an efficient water reservoir under certain conditions.

The framework is applied to two hypothetical intrusion scenarios. The results demonstrate how different assumptions about hydration and composition influence rheology, porosity, and water distribution. Overall, this work provides a numerical basis for exploring hydration-related processes in the Venusian crust.

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1 Introduction

1.1 Background and Motivation

This thesis develops a numerical model to investigate hydration processes in the Venusian crust. Understanding the conditions, feasibility, and possible consequences of these processes has implications for Venus's interior degassing and past evolution. Venus is Earth's closest planetary neighbor, with a minimum distance of approximately 38 million kilometers [11]. Despite this proximity, there have been only a few landing attempts in the 1980s, with only a handful being successful due to Venus's harsh surface conditions: atmospheric pressure 90 times higher than Earth's and surface temperatures at around 740 K due to the greenhouse effect. The landers that successfully touched down survived only between 23 minutes and two hours [11]. Because of this, there is very limited knowledge of Venus and its evolutionary history. This thesis aims to develop a numerical model to investigate how the hydration of Venus's crust might have evolved and what implications certain thermal and hydration structures could have had for the planet's thermal and mechanical evolution.

1.2 Venus Composition Data

Data on the chemical composition of Venus's surface are extremely limited, since only ten landers have successfully landed on Venus. On-site measurements of major element abundances were obtained only by the *Venera 13* and *Venera 14* landers (1982) using X-ray fluorescence (XRF) spectroscopy [16]. The *Vega 1* and *Vega 2* landers (1985) measured potassium (K), uranium (U), and thorium (Th) abundances using gamma-ray spectroscopy [17, 18]. Major oxide compositions at the Vega landing sites were subsequently inferred from these data through petrochemical correlations and comparison with terrestrial analog rocks, rather than measured directly. Due to the extreme surface conditions on Venus and the technological limitations of 1980s instrumentation, all compositional data of the surface are subject to large uncertainties, typically on the order of $\pm$10–30% (1σ). Table 1 summarizes measured and inferred major oxide compositions from these landing sites, including terrestrial analogues for comparison, it has no values of hydration since it was measured by the Venera or VEGA missions.

Table 1: Major oxide compositions of the Venusian surface compared with the Bulk Silicate Earth (BSE). Compositions for Venera 13 and Venera 14 are directly measured by X-ray fluorescence, those for Vega 2 are inferred from gamma-ray K–U–Th measurements, those for Vega 1 is modeled using petrochemical correlations, and BSE represents the terrestrial analogue (adapted from [2, 9]).

Oxide	**Vega 1**[a]	**Vega 2**[b,c]	**Venera 13**[d]	**Venera 14**[d]	**BSE**[g]
SiO_2	50.8	45.6 ± 3.2	45.1 ± 3.0	48.7 ± 3.6	45.0
TiO_2	0.22	0.2 ± 0.1	1.59 ± 0.45	1.25 ± 0.41	0.20
Al_2O_3	17.8	16.0 ± 1.8	15.8 ± 3.0	17.9 ± 2.6	4.45
FeO[e]	7.7	7.7 ± 1.1	9.3 ± 2.2	8.8 ± 1.8	8.05
MnO	0.16	0.14 ± 0.12	0.2 ± 0.1	0.16 ± 0.08	0.14
MgO	12.8	11.5 ± 3.7	11.4 ± 6.2	8.1 ± 3.3	37.8
CaO	8.3	7.5 ± 0.7	7.1 ± 0.96	10.3 ± 1.2	3.55
Na_2O[f]	2.0	2.0	2.0 ± 0.5	2.4 ± 0.4	0.36
K_2O	0.11	0.1 ± 0.08	4.0 ± 0.63	0.2 ± 0.07	0.03
SO_3	–	4.7 ± 1.5	1.62 ± 1.0	0.88 ± 0.77	–
Cl	–	<0.3	<0.3	<0.4	–
Total	100.0	95.4	98.1	98.7	99.6

[a] Modeled basaltic composition consistent with Vega gamma-ray measurements [8].
[b] Major oxide composition inferred for the Vega 2 landing site from gamma-ray spectrometry [17, 18].
[c] Additional trace-element upper limits reported in [17].
[d] Direct XRF measurements from the Venera 13 and Venera 14 landers [16].
[e] All iron reported as total Fe expressed as FeO.
[f] Na_2O calculated following [16, 17].
[g] Bulk Silicate Earth composition from [9], representing the terrestrial analogue.

1.3 Research Objectives

The objective was to determine how different thermal and hydration structures of the Venusian crust affect its evolution. For this purpose, a numerical model was implemented to investigate how density, porosity, and other material properties affect each other under the influence of different hydration levels.

2 Theoretical Framework

This section presents the governing equations for the coupled hydro-thermo-mechanical-chemical system. The formulation follows the continuum approach commonly used in geodynamic models of porous media [5].

2.1 Hydro-Mechanical Model

The hydro-mechanical evolution of the porous medium is described by a coupled system consisting of momentum conservation for the solid matrix, Darcy flow for the pore fluid, and mass conservation equations for both solid and fluid phases.

2.1.1 Momentum Conservation

Assuming creeping flow and neglecting inertial terms, momentum conservation for the solid matrix reduces to the Stokes equations,

$$\nabla \cdot \left[\eta \left(\nabla \boldsymbol{v} + \nabla \boldsymbol{v}^T\right)\right] - \nabla P_t = -\rho \boldsymbol{g}, \tag{1}$$

where $\boldsymbol{v}$ [m·s^{-1}] is the solid velocity, η [Pa·s] is the effective viscosity, P_t [Pa] is the total pressure, ρ [kg·m^{-3}] is the bulk density, and $\boldsymbol{g}$ [m·s^{-2}] is the gravitational acceleration.

2.1.2 Darcy Flow

The relative motion of the pore fluid with respect to the solid matrix is governed by Darcy's law,

$$\boldsymbol{R}\,\boldsymbol{q} + \nabla P_f = \rho_f \boldsymbol{g}, \tag{2}$$

where $\boldsymbol{q}$ [m·s^{-1}] is the Darcy flux, P_f [Pa] is the fluid pressure, ρ_f [kg·m^{-3}] is the fluid density, and $\boldsymbol{R}$ [Pa·s·m^{-2}] $= \mu/k$ is the hydraulic resistance, with μ [Pa·s] denoting fluid viscosity and k [m^2] permeability.

2.1.3 Mass Conservation

Mass conservation is enforced separately for the solid matrix and the pore fluid. The bulk mass balance of the solid phase is given by

$$\nabla \cdot \boldsymbol{v} + \frac{P_t - P_f}{\eta_\phi (1-\phi)} + \beta_d \left(\frac{D_s P_t}{Dt} - K_{bw} \frac{D_f P_f}{Dt} \right) = D_{mp} + \alpha_s \frac{D_s T}{Dt}, \tag{3}$$

where ϕ is porosity (volume fraction of fluid), η_ϕ [Pa·s] is the effective pore viscosity, β_d [Pa^{-1}] is the drained compressibility, K_{bw} is the Biot–Willis coefficient, D_{mp} [s^{-1}] represents mass sources or sinks, and α_s [K^{-1}] is the solid thermal expansion coefficient.

The fluid mass balance equation reads

$$\nabla \cdot \boldsymbol{q} - \frac{P_t - P_f}{\eta_\phi (1-\phi)} - K_{bw}\beta_d \left(\frac{D_s P_t}{Dt} - \frac{1}{K_{sk}} \frac{D_f P_f}{Dt} \right) = \phi \left(\alpha_f - \alpha_s \right) \frac{D_s T}{Dt}, \quad (4)$$

where α_f [K^{-1}] is the fluid thermal expansion coefficient and K_{sk} is the inverse Skempton coefficient.

2.1.4 Porosity evolution

Porosity is advected with the solid phase and evolves in response to volumetric compaction or dilation of the porous matrix. Its evolution is written in terms of the material derivative following the solid velocity,

$$\frac{D_s}{Dt} \ln\left(\frac{1-\phi}{\phi} \right) \; = \; A_\phi, \quad (5)$$

where A_ϕ is the local compaction–dilation rate, which depends on the mechanical state of the solid–fluid system.

Equivalently, the porosity evolution equation can be written as

$$\frac{D_s \phi}{Dt} = -\phi(1-\phi)\, A_\phi, \quad (6)$$

showing that positive A_ϕ corresponds to compaction (porosity reduction), whereas negative A_ϕ leads to dilation.

2.1.5 Viscosity evolution

The effective solid viscosity is computed assuming a composite viscous rheology in which diffusion creep and dislocation creep act in parallel. The resulting viscosity is given by the harmonic sum

$$\eta_{\mathrm{eff}}^{-1} = \eta_{\mathrm{dif}}^{-1} + \eta_{\mathrm{dis}}^{-1}, \quad (7)$$

where η_{dif} is the diffusion creep viscosity and η_{dis} is the non-Newtonian dislocation creep viscosity.

Dislocation creep follows a power-law relation of the form

$$\dot{\varepsilon}_{II} = A_D\, \sigma_{II}^n \exp\left(-\frac{E_a + PV_a}{RT} \right), \quad (8)$$

where $\dot{\varepsilon}_{II}$ and σ_{II} are the second invariants of strain rate and deviatoric stress, respectively, A_D is the pre-exponential factor, n the stress exponent, E_a the activation energy, V_a the activation volume, R the gas constant, and T the temperature.

2.2 Thermal Model

The thermal evolution of the system is governed by conservation of energy, accounting for conductive heat transport and internal heat production. Temperature is solved on the same Eulerian grid as the mechanical variables using an implicit finite-difference method. scheme [5].

2.2.1 Energy Conservation

Energy conservation for the porous medium is written as

$$\rho c_p \frac{\partial T}{\partial t} = -\nabla \cdot \boldsymbol{q}_T + H_r + H_s + H_a + H_p, \tag{9}$$

where T [K] is temperature, ρc_p [$\mathrm{J{\cdot}m^{-3}{\cdot}K^{-1}}$] is the volumetric heat capacity, and $\boldsymbol{q}_T$ [$\mathrm{W{\cdot}m^{-2}}$] is the conductive heat flux.

The conductive heat flux is given by Fourier's law,

$$\boldsymbol{q}_T = -\boldsymbol{K}\nabla T, \tag{10}$$

where $\boldsymbol{K}$ [$\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$] is the (possibly anisotropic) thermal conductivity tensor.

2.2.2 Heat Sources

The total volumetric heat production consists of several contributions,

$$H = H_r + H_s + H_a + H_p, \tag{11}$$

where H_r [$\mathrm{W{\cdot}m^{-3}}$] denotes radiogenic heat production, H_s [$\mathrm{W{\cdot}m^{-3}}$] is shear (viscous) heating, H_a [$\mathrm{W{\cdot}m^{-3}}$] represents adiabatic heating, and H_p [$\mathrm{W{\cdot}m^{-3}}$] accounts for additional pressure-related or dissipative heat sources included in the model.

2.3 Chemical Model

The chemical evolution is represented by the transport and partitioning of a single volatile component (H_2O) between (i) free pore fluid and (ii) structurally bound water in the solid. Free water is transported relative to the solid by Darcy flow, whereas bound water is immobile and adjusts toward local thermodynamic equilibrium as a function of pressure and temperature through hydration–dehydration reactions.

2.3.1 Water Transport

The molar concentration of free porous water per bulk volume, n_f [$\mathrm{mol\,m^{-3}}$], evolves according to

$$\frac{\partial n_f}{\partial t} = -\frac{1}{M_f}\nabla \cdot (\rho_w \boldsymbol{q}) \; + \; R, \tag{12}$$

where M_f [$\mathrm{kg\,mol^{-1}}$] is the molar mass of water, ρ_w [$\mathrm{kg\,m^{-3}}$] is the pore fluid density, $\boldsymbol{q}$ [$\mathrm{m\,s^{-1}}$] is the Darcy flux, and R [$\mathrm{mol\,m^{-3}\,s^{-1}}$] represents the net production or consumption of free water by hydration–dehydration reactions.

2.3.2 Hydration–Dehydration (Equilibrium Partitioning)

The equilibrium bound-water fraction in the solid, $X^{\mathrm{sol}}_{H,\mathrm{eq}}$, is prescribed from thermodynamic data as a function of pressure, temperature, and total hydration,

$$X^{\mathrm{sol}}_{H,\mathrm{eq}} = X^{\mathrm{sol}}_{H_2O}(P, T, X^{\mathrm{tot}}_H). \tag{13}$$

Let X^{sol}_H denote the current bound-water fraction in the solid and n_s the molar amount of dry solid per bulk volume. The proposed change in solid hydration is

$$\Delta X^{\mathrm{sol}}_H = X^{\mathrm{sol}}_{H,\mathrm{eq}} - X^{\mathrm{sol}}_H, \tag{14}$$

subject to the availability of free water,

$$\Delta X^{\mathrm{sol}}_H \leq \frac{n_f}{n_s}. \tag{15}$$

Water is exchanged between free and bound reservoirs while conserving total water:

$$n_f^{\,n+1} = n_f^* - n_s\,\Delta X^{\mathrm{sol}}_H, \qquad X_H^{\mathrm{sol},\,n+1} = X_H^{\mathrm{sol},\,n} + \Delta X^{\mathrm{sol}}_H, \tag{16}$$

where superscripts n and $n+1$ correspond to discrete time levels before and after the update and n_f^* denotes the free-water concentration after the transport step. Bound water increases during hydration ($\Delta X^{\mathrm{sol}}_H > 0$) at the expense of free water and decreases during dehydration.

The total hydration of the solid–fluid mixture is defined as

$$X^{\mathrm{tot}}_H = \frac{n_s X^{\mathrm{sol}}_H + n_f}{n_s + n_f}, \tag{17}$$

which is conserved during the reaction step and evolves only through advective transport and volumetric deformation.

Over a time step Δt, the effective reaction rate appearing in the free-water balance is given by

$$R = -\frac{n_s\,\Delta X^{\mathrm{sol}}_H}{\Delta t}, \tag{18}$$

such that hydration ($\Delta X^{\mathrm{sol}}_H > 0$) consumes free water ($R < 0$), whereas dehydration produces free water ($R > 0$).

2.4 Thermodynamic Model

Phase equilibria and thermodynamic properties are computed using Gibbs free-energy minimization at fixed pressure, temperature, hydration, and bulk composition. The formulation is based on equilibrium thermodynamics and allows for partitioning of water between solid phases and a free fluid phase.

2.4.1 Gibbs Free Energy Minimization

At given pressure P and temperature T, the stable phase assemblage minimizes the total Gibbs free energy of the system,

$$G = \sum_{i=1}^{N_{\mathrm{ph}}} n_i \, \bar{G}_i(P, T, \mathbf{X}_i), \tag{19}$$

where n_i is the amount (in moles) of phase i, $\bar{G}_i$ is its molar Gibbs free energy, and $\mathbf{X}_i$ denotes the internal composition of the phase.

For each phase, the molar Gibbs free energy is expressed as

$$\bar{G}_i = \bar{G}_i^0(P, T) + RT \sum_j X_{ij} \ln a_{ij} + G_i^{\mathrm{ex}}, \tag{20}$$

where $\bar{G}_i^0$ is the reference-state free energy, X_{ij} and a_{ij} are the mole fraction and activity of component j in phase i, respectively, and G_i^{ex} represents non-ideal excess mixing contributions.

2.4.2 Bulk Composition Constraints

The Gibbs free energy minimization is subject to bulk composition constraints expressed in terms of oxide components,

$$\sum_{i=1}^{N_{\mathrm{ph}}} n_i \, s_{ik} = B_k, \qquad k = 1, \ldots, N_{\mathrm{comp}}, \tag{21}$$

where s_{ik} is the stoichiometric amount of component k in phase i and B_k is the bulk amount of component k. Water (H_2O) is treated as an independent thermodynamic component and may be incorporated into solid phases or stabilized as a separate fluid phase, depending on pressure, temperature, and bulk hydration.

2.4.3 Phase Equilibria

Phase equilibria are defined by the equality of chemical potentials of all components between coexisting phases,

$$\mu_k^{\alpha}(P, T) = \mu_k^{\beta}(P, T), \tag{22}$$

where $\mu_k = \partial G / \partial n_k$ is the chemical potential of component k, and α and β denote coexisting phases. Phase assemblages, phase compositions, and phase proportions are determined self-consistently by Gibbs' free energy minimization.

2.4.4 Thermodynamic Properties of Solids

Once the equilibrium phase assemblage is obtained, the bulk thermodynamic properties of the solid system are computed as phase-fraction-weighted averages. Density is given by

$$\rho = \frac{\sum_i n_i M_i}{\sum_i n_i \bar{V}_i}, \tag{23}$$

where $\bar{V}_i = \left(\partial \bar{G}_i / \partial P\right)_T$ is the molar volume of phase i, M_i is the molar mass of phase i, and n_i is the number of moles of phase i. The total volume is

$$V = \sum_i n_i \bar{V}_i. \tag{24}$$

Enthalpy is obtained from

$$H = \sum_i n_i \bar{H}_i, \tag{25}$$

with

$$\bar{H}_i = \bar{G}_i - T \left(\frac{\partial \bar{G}_i}{\partial T}\right)_P. \tag{26}$$

The heat capacity at constant pressure is computed as

$$C_p = \sum_i n_i \bar{C}_{p,i}, \qquad \bar{C}_{p,i} = -T \left(\frac{\partial^2 \bar{G}_i}{\partial T^2}\right)_P. \tag{27}$$

Thermal expansivity and compressibility are obtained from the second derivatives of the Gibbs free energy,

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P, \qquad \beta = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T. \tag{28}$$

2.4.5 Fluid Thermodynamics

Thermodynamic properties of the H_2O fluid phase are computed independently using the Pitzer–Sterner equation of state [12]. For the fluid phase, the Gibbs free energy is expressed as a function of pressure and temperature,

$$G_f = G_f(P, T), \tag{29}$$

from which density, enthalpy, heat capacity, thermal expansivity, and compressibility are derived consistently using standard thermodynamic relations,

$$V_f = \left(\frac{\partial G_f}{\partial P}\right)_T, \qquad H_f = G_f - T\left(\frac{\partial G_f}{\partial T}\right)_P, \tag{30}$$

$$C_{p,f} = -T\left(\frac{\partial^2 G_f}{\partial T^2}\right)_P, \qquad \alpha_f = \frac{1}{V_f}\left(\frac{\partial V_f}{\partial T}\right)_P, \qquad \beta_f = -\frac{1}{V_f}\left(\frac{\partial V_f}{\partial P}\right)_T. \tag{31}$$

3 Numerical Methods

3.1 Numerical Implementation

The numerical implementation follows the approach described in [5]. The code is based on an HTMC framework with a coupled solid–liquid system including an aqueous fluid phase [19]. The detailed implementation of the thermodynamic database computation is provided in [10], and the HTMC model implementation is documented in [7].

Step-by-step algorithm

1. **Compute database 3.2:** build/load the required material/property tables.
2. **Initialize 4.5:** set initial fields, markers, boundary conditions, and parameters.
3. **Timestep 3.10.3:** advance time and update timestep-dependent quantities.
4. **HTM iteration:** iterate the coupled hydro-thermo-mechanical system.
5. **Update 3.6 and interpolate 3.4:** lookup thermodynamic properties from database at markers, then interpolate η, ρ, ϕ, k to grid.
6. **Stokes solve 3.5.1:** Composing matrices for Stokes and continuity equations.
7. **Temperature solve 3.5.2:** Composing matrices to solve for T.
8. **Convergence check:** if P_f not converged, repeat HTM iteration.
9. **Update fields 3.7, 3.8, 3.9:** update ϕ, hydration, η.
10. **Markers 3.4:** move and add markers as needed.
11. **Plot:** Plot fields.
12. **End check:** if time not ended, return to timestep.

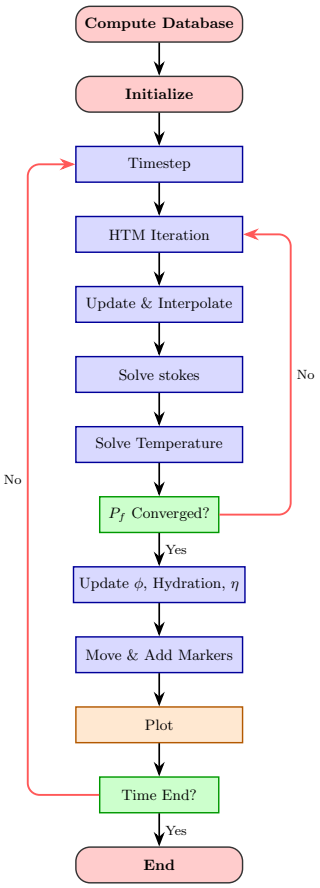


Figure 1: Flowchart of the hydro-thermo-mechanical-chemical algorithm with a step-by-step description.

3.2 Thermodynamic Database Construction

Phase equilibria for a selected Venus composition 2 are computed as a function of pressure, temperature, and hydration (mol fraction) in a selected limited range. Solid properties of the equilibrated oxides were computed by the MAGEMin minimization [14] in Julia following the theory from Section 2.4, while properties of the fluid (water) were computed by the Pitzer and Sterner EoS solver [12] described in Section 2.4.5& 3.10.2, the detailed implementation is in [10]. The database consists of the following inputs and outputs:

- **Independent state variables (database inputs)**
  - Pressure (P)
  - Temperature (T)
  - Hydration (mole fraction, $x_{\mathrm{H_2O}}$)
- **Dependent variables (database outputs)**
  - **Thermodynamic properties**
    - Density (ρ)
    - Enthalpy (H^0)
    - Heat capacity (C_p)
    - Molar volume (V)
  - **Phase equilibrium variables**
    - Solid phase composition ($x^{\mathrm{solid}}_{\mathrm{H_2O}}$)
  - **Thermodynamic response functions**
    - Thermal expansivity (α)
    - Compressibility (β)

3.3 Staggered Grid

A fully staggered grid with ghost nodes was used according to [5]. The layout can be seen in Figure 2.

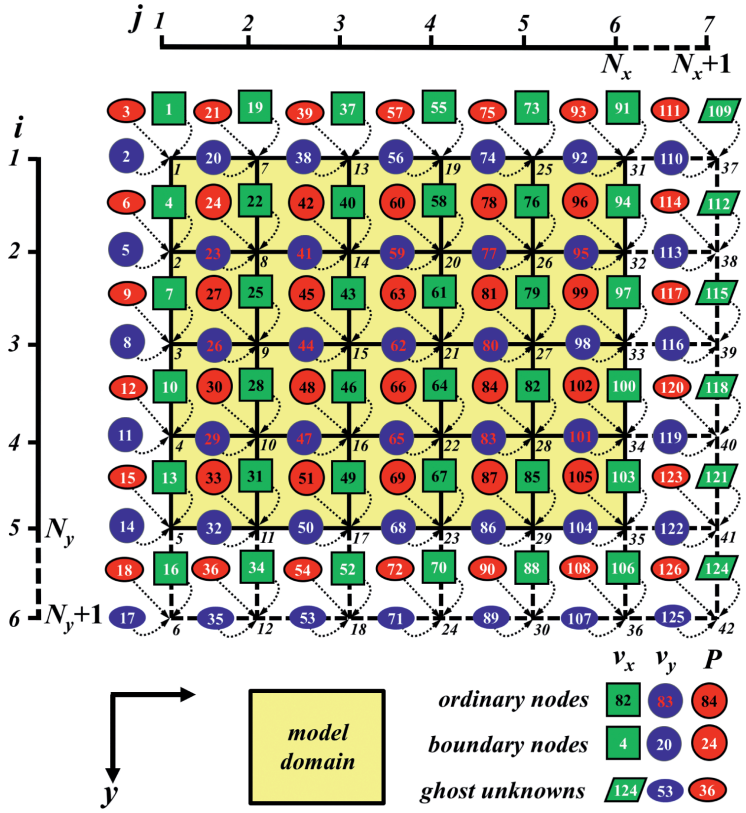


Figure 2: Staggered grid layout [5].

The different node types (P-nodes, v_x-nodes, v_y-nodes, and basic nodes) hold different variables as follows:

v_x-nodes (horizontal velocity nodes): v_x, q_{Dx}, ρ_x, k_x, ϕ_x, R_x

v_y-nodes (vertical velocity nodes): v_y, q_{Dy}, ρ_y, k_y, ϕ_y, R_y

Basic nodes (regular grid intersections): η_s, $\dot{\varepsilon}_{xy}$, σ'_{xy}

P-nodes (pressure nodes): P_t, P_f, η_t, η_ϕ, $\dot{\varepsilon}_{xx}$, $\dot{\varepsilon}_{yy}$, σ'_{xx}, σ'_{yy}, T, α, , ρ, H_s, H_a, H_r, ρC_p, ϕ, X_H

Some variables are stored on different node types because they are needed at different locations for different equations.

3.4 Marker-in-Cell

The Marker-in-Cell method, according to [5] was used to reduce numerical diffusion by combining Eulerian and Lagrangian advection schemes. The so-called markers are Lagrangian advection points that are advected using fourth-order Runge-Kutta integration according to the velocity field. Physical properties of the Lagrangian markers are interpolated to the Eulerian grid nodes with a weighted distance average, such that these properties can be used on the Eulerian grid for stencil operations or plotting.

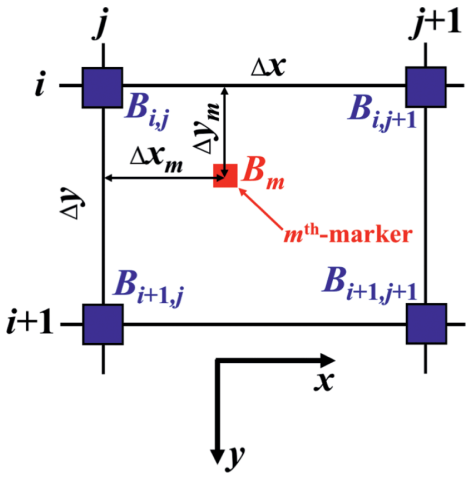


Figure 3: Marker to node interpolation [5].

3.5 Finite Difference Stencils

The stencils are based on [5]. These stencils are then applied to the full staggered grid from Figure 2 to set up the linear system of equations. One linear system of equations for the Hydro-Mechanical Stencils 3.5.1 and one for the Thermal Stencil 3.5.2. These systems of equations are then solved 3.10.1 to obtain the solution.

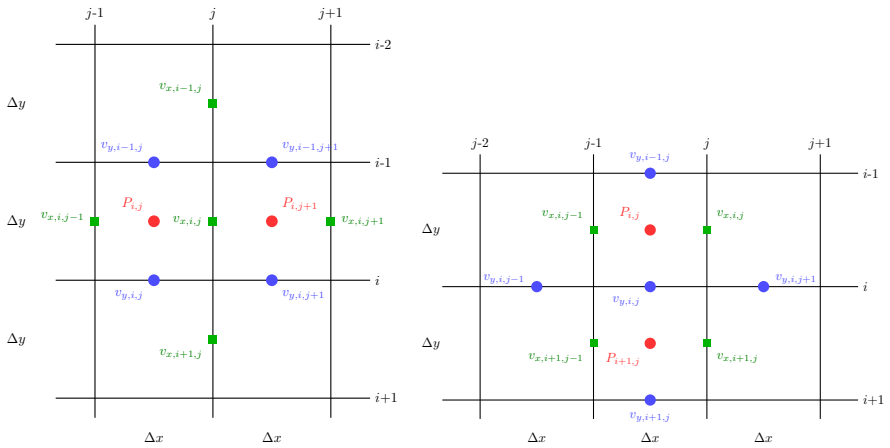
3.5.1 Hydro-Mechanical Stencils

The governing equations are discretized using second-order centered finite difference stencils on the full staggered grid [5].

Momentum Conservation (Stokes Equations) The x-component of momentum conservation is discretized at v_x-nodes, and the y-component at v_y-nodes (Figure 4):

$$
\begin{aligned}
&\frac{\eta_{p,i,j}(v_{x,i,j-1} - v_{x,i,j}) + \eta_{p,i,j+1}(v_{x,i,j+1} - v_{x,i,j})}{\Delta x^2} \\
&+ \frac{\eta_{i-1,j}(v_{x,i-1,j} - v_{x,i,j}) + \eta_{i,j}(v_{x,i+1,j} - v_{x,i,j})}{\Delta y^2} \\
&+ \frac{1}{\Delta x \Delta y}\Big(\eta_{p,i,j}(v_{y,i,j} - v_{y,i-1,j}) - \eta_{p,i,j+1}(v_{y,i,j+1} - v_{y,i-1,j+1}) \\
&\qquad + \eta_{i,j}(v_{y,i+1,j} - v_{y,i,j}) - \eta_{i-1,j}(v_{y,i,j} - v_{y,i-1,j})\Big) \\
&- \frac{P_{i,j+1} - P_{i,j}}{\Delta x} = -\rho_{i,j} g_x.
\end{aligned} \tag{32}
$$

$$
\begin{aligned}
&\frac{\eta_{i,j-1}(v_{y,i,j-1} - v_{y,i,j}) + \eta_{i,j}(v_{y,i,j+1} - v_{y,i,j})}{\Delta x^2} \\
&+ \frac{\eta_{p,i,j}(v_{y,i-1,j} - v_{y,i,j}) + \eta_{p,i+1,j}(v_{y,i+1,j} - v_{y,i,j})}{\Delta y^2} \\
&+ \frac{1}{\Delta x \Delta y}\Big(\eta_{p,i,j}(v_{x,i,j} - v_{x,i-1,j}) - \eta_{p,i+1,j}(v_{x,i,j+1} - v_{x,i-1,j+1}) \\
&\qquad + \eta_{i,j}(v_{x,i,j+1} - v_{x,i,j}) - \eta_{i,j-1}(v_{x,i-1,j+1} - v_{x,i-1,j})\Big) \\
&- \frac{P_{i+1,j} - P_{i,j}}{\Delta y} = -\rho_{i,j} g_y.
\end{aligned} \tag{33}
$$



(a) x-Stokes equation stencil (b) y-Stokes equation stencil

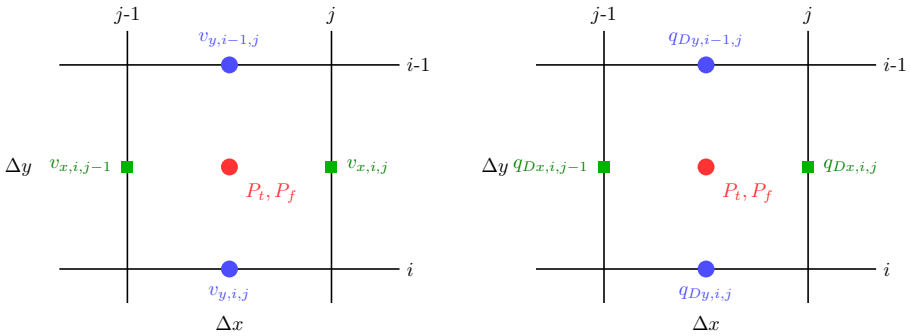
Figure 4: Stencils used for the discretization of the Stokes equations on a 2D regular staggered grid. Velocity components (v_x, v_y) and pressure (P) are the unknowns shown.

Mass Conservation (Continuity Equations) The total mass balance is enforced at pressure nodes (Figure 5a):

$$\begin{aligned}\frac{v_{x,i,j} - v_{x,i,j-1}}{\Delta x} + \frac{v_{y,i,j} - v_{y,i-1,j}}{\Delta y} + \frac{P_t - P_f}{\eta_\phi(1-\phi)} \\ + \beta_d \left(\frac{D_s P_t}{Dt} - K_{bw} \frac{D_f P_f}{Dt} \right) = D_{mp} + \alpha_s \frac{D_s T}{Dt}.\end{aligned} \tag{34}$$

The fluid mass balance is also enforced at pressure nodes (Figure 5b):

$$\begin{aligned}\frac{q_{Dx,i,j} - q_{Dx,i,j-1}}{\Delta x} + \frac{q_{Dy,i,j} - q_{Dy,i-1,j}}{\Delta y} - \frac{P_t - P_f}{\eta_\phi(1-\phi)} \\ - K_{bw}\beta_d \left(\frac{D_s P_t}{Dt} - K_{sk}^{-1} \frac{D_f P_f}{Dt} \right) = \phi \left(\alpha_f \frac{D_f T_f}{Dt} - \alpha_s \frac{D_s T}{Dt} \right).\end{aligned} \tag{35}$$



(a) Velocities (v_x, v_y), total pressure (P_t), and fluid pressure (P_f) are the unknowns shown

(b) Darcy fluxes (q_{Dx}, q_{Dy}), total pressure (P_t), and fluid pressure (P_f) are the unknowns shown

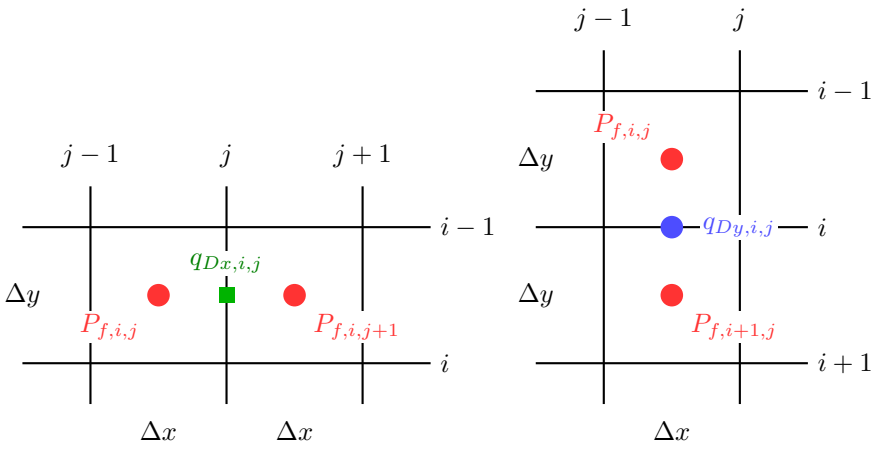
Figure 5: Five-point stencils for total mass and fluid mass conservation.

Darcy Flow Equations The x-component of Darcy flux is computed at q_{Dx}-nodes (same grid as v_x), and the y-component at q_{Dy}-nodes (same grid as v_y):

$$R_x q_{Dx,i,j} + \frac{P_{f,i,j+1} - P_{f,i,j}}{\Delta x} = \rho_{f,i,j} g_x, \tag{36}$$

$$R_y q_{Dy,i,j} + \frac{P_{f,i+1,j} - P_{f,i,j}}{\Delta y} = \rho_{f,i,j} g_y, \tag{37}$$

where $R_x = \mu/k$ and $R_y = \mu/k$ are the hydraulic resistances (Figure 6).



(a) x-Darcy equation stencil (b) y-Darcy equation stencil

Figure 6: Stencils used for the discretization of the Darcy flow equations on a 2D staggered grid. Darcy fluxes (q_{Dx}, q_{Dy}) and fluid pressures (P_f) are the unknowns shown.

3.5.2 Thermal Stencil

Thermal Equation The thermal model uses the discretized heat equation $\rho C_p \frac{\partial T}{\partial t} = -\frac{\partial q_x}{\partial x} - \frac{\partial q_y}{\partial y} + H_r + H_s + H_a + H_p$, which is solved separately from the hydro-mechanical system. Temperature is solved at pressure nodes using a five-point stencil:

$$\begin{aligned} \rho C_p \frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t} = & \frac{K_{x,i,j-1}(T_{i,j-1}^{n} - T_{i,j}^{n}) + K_{x,i,j}(T_{i,j+1}^{n} - T_{i,j}^{n})}{\Delta x^2} \\ & + \frac{K_{y,i-1,j}(T_{i-1,j}^{n} - T_{i,j}^{n}) + K_{y,i,j}(T_{i+1,j}^{n} - T_{i,j}^{n})}{\Delta y^2} \\ & + H_r + H_s + H_a + H_p, \end{aligned} \tag{38}$$

Where $T_{i,j}^{n}$ and $T_{i,j}^{n+1}$ are temperatures at the old and new time levels, K_x, K_y are thermal conductivities at cell faces, and H_r, H_s, H_a, H_p represent radiogenic, shear, adiabatic, and other heat production terms (Figure 7).

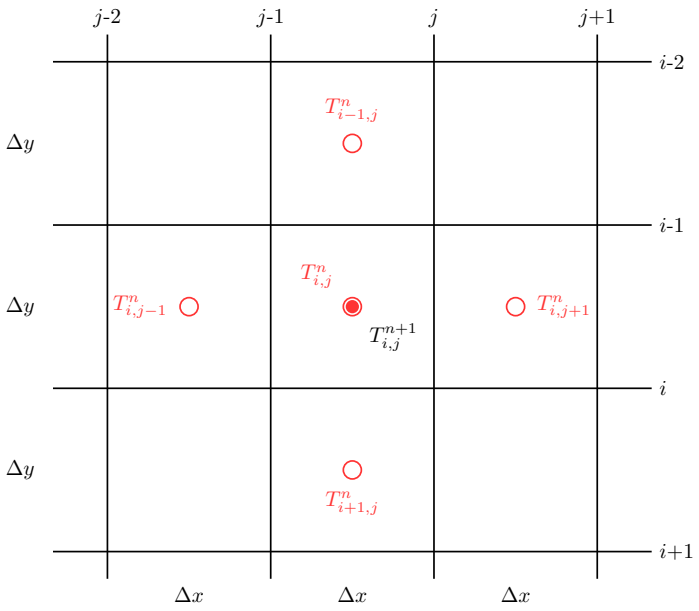


Figure 7: Five-point stencil for the thermal diffusion equation. The filled node represents the unknown temperature $T^{n+1}_{i,j}$, while hollow nodes denote known neighboring temperatures used in the stencil.

3.6 Thermodynamic Update

The dependent state variables 3.2 are interpolated trilinearly using the three independent state variables 3.2: pressure (P), temperature (T), and hydration (mole fraction, $x_{\mathrm{H_2O}}$) from the precomputed database 3.2. Limitations of this approach are listed in Section 5.3.1.

3.7 Porosity Update

The numerical update of porosity follows the physical evolution law described in Section 2.1.4. The compaction–dilation rate A_ϕ entering the porosity evolution equation is defined on the Eulerian grid. For each marker, A_ϕ is obtained by bilinear interpolation from the surrounding grid nodes.

In practice, we distinguish between a *compositional porosity* $\phi_{\mathrm{com}} \in (0, 1)$, which is the quantity evolved by the porosity update, and a *mechanical porosity* $\phi \in (\phi_{\min}, \phi_{\max})$, which is used in the governing equations. Quantities with superscripts n and $n+1$ correspond to discrete time levels before and after the update, respectively.

$$\chi^n = \ln\left(\frac{1-\phi^n}{\phi^n}\right), \tag{39}$$

which is advanced explicitly in time as

$$\chi^{n+1} = \chi^n + A_\phi \, \Delta t. \tag{40}$$

The updated compositional porosity is then recovered by,

$$\phi_{\text{com}}^{n+1} = \frac{1}{(1-\phi^n)\,\exp(A_\phi \Delta t) + \phi^n}. \tag{41}$$

The mechanical porosity is finally obtained by bounding the compositional porosity within prescribed limits,

$$\phi^{n+1} = \min\!\big(\max\!\big(\phi_{\text{com}}^{n+1}, \phi_{\min}\big), \phi_{\max}\big), \tag{42}$$

where $\phi_{\min}$ and $\phi_{\max}$ are numerical parameters.

This separation is required because porosity enters several relations of the hydro-mechanical system in a non-linear manner, for example through terms proportional to $(1-\phi)^{-1}$. Allowing porosity values close to 0 or 1 would lead to numerical issues; more details regarding that are explained in Section 5.3.

3.8 Hydration Update

After the porosity update, the hydration state of each marker is updated by explicitly accounting for (i) Darcy transport of free pore water and (ii) local partitioning of water between the free fluid and solid-bound reservoirs.

First, fluid pressure, temperature, Darcy mass-flux divergence, fluid density, and solid density are interpolated from the Eulerian grid to each Lagrangian marker using bilinear interpolation. The dry solid density is then computed from the bulk solid density by removing the contribution of structurally bound water,

$$\rho_{\text{dry}} = \frac{\rho_{\text{solid}}}{1 + X_{\text{H}}^{\text{solid}}\,(M_f / M_{\text{dry}})}, \tag{43}$$

where M_f and M_{dry} are the molar masses of water and dry solid, respectively.

The molar amount of dry solid per bulk volume,

$$n_s = \frac{(1-\phi)\,\rho_{\text{dry}}}{M_{\text{dry}}}, \tag{44}$$

is then used to compute the molar concentration of free water per bulk volume, n_f, from the marker total hydration $X_{\text{H}}^{\text{tot}}$ and the current solid hydration $X_{\text{H}}^{\text{solid}}$,

$$n_f = n_s \, \frac{X_{\text{H}}^{\text{tot}} - X_{\text{H}}^{\text{solid}}}{1 - X_{\text{H}}^{\text{tot}}}. \tag{45}$$

Negative values of n_f are prevented by construction.

Transport of free water is computed explicitly using the discretized form of Eq. (8). The Darcy divergence provided by the flow solver, $\nabla \cdot (\rho_w \mathbf{q})$, is converted to a molar rate and applied as

$$\Delta n_f = -\frac{1}{M_f} \left[\nabla \cdot (\rho_w \mathbf{q})\right] \Delta t, \tag{46}$$

where Δt is the time step. Removal of free water is limited such that n_f remains non-negative.

After free-water transport, the total hydration of the marker is recomputed as

$$X_{\mathrm{H}}^{\mathrm{tot}} = \frac{n_s X_{\mathrm{H}}^{\mathrm{solid}} + n_f}{n_s + n_f}. \tag{47}$$

At the updated pressure, temperature, and total hydration, the equilibrium solid hydration $X_{\mathrm{H}}^{\mathrm{solid,eq}}$ is obtained by trilinear interpolation of the precomputed thermodynamic database. The change in bound water implied by equilibrium,

$$\Delta X_{\mathrm{H}}^{\mathrm{solid}} = X_{\mathrm{H}}^{\mathrm{solid,eq}} - X_{\mathrm{H}}^{\mathrm{solid}}, \tag{48}$$

is applied subject to water availability in the pore space,

$$\Delta X_{\mathrm{H}}^{\mathrm{solid}} \leq \frac{n_f}{n_s}. \tag{49}$$

The exchange between free and bound water is then enforced as (superscripts n and $n+1$ correspond to discrete time levels before and after the update)

$$n_f^{n+1} = n_f^n - n_s\, \Delta X_{\mathrm{H}}^{\mathrm{solid}}, \qquad X_{\mathrm{H}}^{\mathrm{solid},\,n+1} = X_{\mathrm{H}}^{\mathrm{solid},\,n} + \Delta X_{\mathrm{H}}^{\mathrm{solid}}. \tag{50}$$

ensuring conservation of total water at the marker scale. Finally, the total hydration is recomputed using Eq. (12), and all hydration variables are bounded to physically meaningful values.

3.9 Viscosity Update

Viscosity is evaluated on Lagrangian markers using strain rate, pressure, and temperature interpolated from the Eulerian grid. The second invariant of the strain-rate tensor is computed from the normal and shear components,

$$\dot{\varepsilon}_{II} = \left(\dot{\varepsilon}_{xx}^2 + \dot{\varepsilon}_{xy}^2\right)^{1/2}. \tag{51}$$

For dislocation creep, the effective viscosity is obtained implicitly from the power-law relation by iteratively solving for consistency between stress and strain rate. A bisection method is employed to determine the viscosity within prescribed bounds $[\eta_{\min}, \eta_{\max}]$. Diffusion and dislocation creep viscosities are then combined using the harmonic average to obtain the effective marker viscosity.

The rheological parameters entering the power-law formulation (pre-exponential factor, stress exponent, and activation energy) are material-dependent and are summarized in Section 4.2.1 and Table 3.

This procedure ensures numerical stability while preserving the non-linear rheological behavior implied by the experimental flow laws. Dislocation creep follows a power-law rheology relating the second invariants of stress and strain rate,

$$\dot{\varepsilon}_{II} = A_D\, \sigma_{II}^n \exp\!\left(-\frac{E_a + PV_a}{RT}\right), \tag{52}$$

with $\sigma_{II} = 2\eta\dot{\varepsilon}_{II}$. Diffusion and dislocation creep act in parallel, such that the effective viscosity is given by the harmonic sum

$$\eta_{\mathrm{eff}}^{-1} = \eta_{\mathrm{dif}}^{-1} + \eta_{\mathrm{dis}}^{-1}. \tag{53}$$

3.10 Solution Algorithm

3.10.1 Sparse Matrix Solver

To solve the linear systems arising from the Stokes and temperature equations, we use MATLAB's built-in matrix division operator,

$$S = R \backslash L,$$

which invokes optimized sparse direct solvers based on matrix structure rather than explicitly forming the matrix inverse [20].

3.10.2 $\mathrm{H_2O}$ Equation of State Solver

Thermodynamic properties of the $\mathrm{H_2O}$ fluid are evaluated numerically using the Pitzer–Sterner [12] equation of state. The EOS is implemented in the form $P(\rho, T)$, where the temperature-dependent coefficients are obtained by multiplying the published coefficient matrix by a polynomial basis in temperature. For a given pressure and temperature, the molar density is recovered by numerically inverting the EOS using a bisection scheme over a fixed number of iterations.

Pressure contributions to the Gibbs free energy are accounted for by numerical evaluation of the integral

$$\int_{P_r}^{P} V(P', T)\,\mathrm{d}P', \tag{54}$$

where the volume is obtained from the EOS at each quadrature point. Reference-state thermal contributions to enthalpy and entropy are computed using analytic heat-capacity integrals following the Holland–Powell parameterization. The Gibbs free energy is then assembled as a function of pressure and temperature.

Fluid enthalpy is calculated from the Gibbs free energy using the thermodynamic relation

$$H_f = -T^2 \left(\frac{\partial (G_f/T)}{\partial T} \right)_P, \tag{55}$$

where the temperature derivative is evaluated numerically using automatic differentiation. Density, thermal expansivity, and compressibility are obtained consistently from finite-difference derivatives on the computed pressure–temperature grid.

3.10.3 Timestep Control

The numerical timestep is constrained by three independent stability and accuracy criteria. Violation of any of these constraints may lead to numerical instability or physically corrupted results.

1. Solid velocity: The timestep is limited by the CFL (Courant-Friedrichs-Lewy) condition. The CFL condition requires that, during a single timestep, information must not propagate over more than a prescribed fraction of the spatial discretization length. For a Cartesian grid, this condition can be expressed as

$$\Delta t \leq C_{\mathrm{CFL}} \min\left(\frac{\Delta x}{|v_x|_{\max}}, \frac{\Delta y}{|v_y|_{\max}}\right), \tag{56}$$

where Δx and Δy are the grid spacings, $|v_x|_{\max}$ and $|v_y|_{\max}$ are the maximum absolute solid velocities in the respective directions, and $C_{\mathrm{CFL}} < 1$ is a user-defined Courant number[1].

2. Composition change: The timestep is constrained by the maximum allowable change in chemical composition per timestep. In this study, the change in water content is limited by a prescribed maximum molar variation, ensuring gradual compositional evolution and preventing unphysical overshooting caused by rapid reaction rates.

3. Porosity change: The timestep is further restricted by the maximum permissible change in porosity per timestep. This constraint guarantees numerical stability of the coupled hydro-mechanical system and avoids abrupt porosity variations that could destabilize the pressure–permeability feedback.

4 Model Setup

4.1 Venus Compositions

Due to the limited data on Venus' crust, the model is capable of evaluating different crustal compositions of Venus ,all summarized in Table 2. There are site-specific compositions from the landers *Venera 13* and *Venera 14* which are based on direct X-ray fluorescence measurements of major element abundances [16]. Inferred and modelled composition from the landers *Vega 1* and *Vega 2*. In addition to the site specific composition from Table 1 there is a representative Venus basalt model from [15], which was constructed to be consistent with the range of major-element abundances measured or inferred by the Venera and Vega landers [16, 17, 18] and with experimental constraints on Venus magmatism derived from terrestrial analogs studies [21, 2]. For comparison with a terrestrial reference composition, the Bulk Silicate Earth (BSE) is also included as a terrestrial analogue [9].

Table 2: Venus compositions used as inputs to compute the thermodynamic database. Lander-based compositions are reproduced from Table 1, the representative Venus basalt is a synthetic modeling composition adopted from [15], and the Bulk Silicate Earth (BSE) is included as a terrestrial analogue [9].

Oxide (wt.%)	**Venera 13**	**Venera 14**	**Vega 2**	**Vega 1**	**Venus Basalt**[a]	**BSE**[d]
SiO_2	45.1	48.7	45.6	50.8	48.7	45.0
TiO_2	1.59	1.25	0.20	0.22	1.30	0.20
Al_2O_3	15.8	17.9	16.0	17.8	17.9	4.45
FeO[b]	9.3	8.8	7.7	7.7	7.92	8.05
Fe_2O_3[c]	–	–	–	–	0.98	–
MgO	11.4	8.1	11.5	12.8	8.1	37.8
CaO	7.1	10.3	7.5	8.3	10.3	3.55
Na_2O	2.0	2.4	2.0	2.0	2.4	0.36
K_2O	4.0	0.2	0.1	0.11	0.2	0.03
Total	98.1	98.7	95.4	100.0	97.8	99.6

[a] Representative synthetic Venus basalt adopted from [15], constructed to be consistent with Venus lander constraints and terrestrial experimental analogs [21, 2].
[b] Total iron reported as FeO for lander-based compositions.
[c] Iron oxidation state assumed for the representative Venus basalt only.
[d] Bulk Silicate Earth composition from [9], representing the terrestrial analogue.

4.2 Material Properties

4.2.1 Viscosity

Viscous rheology is described using experimentally derived, non-Newtonian power-law flow laws following [5]. We implement three of the rheological laws listed in Table 21.1 of [5], which were originally compiled from laboratory deformation experiments summarized by [13].

In the numerical model of this thesis, three flow laws are employed: plagioclase (An_{75}), diabase, and mafic granulite. Each material is characterized by a pre-exponential constant A_D [$MPa^{-n}\,s^{-1}$], stress exponent n, and activation energy E_a [$kJ\,mol^{-1}$]. The adopted parameters are summarized in Table 3.

Table 3: Rheological flow-law parameters used in the numerical model. Values from Table 21.1 of [5], originally compiled from [13].

Material	A_D [$MPa^{-n}\ s^{-1}$]	n	E_a [kJ mol^{-1}]
Plagioclase (An_{75})	3.3×10^{-4}	3.2	238
Diabase	2.0×10^{-4}	3.4	260
Mafic granulite	1.4×10^{4}	4.2	445

4.2.2 Density

Density is dependent on pressure, temperature, and hydration. The phase equilibria were computed using MAGEMin [14], dependent on the different compositions of Venus (Table 2).

4.3 Timestep Parameters

Time integration uses an adaptive timestep Δt, initialized as $\Delta t_0 = 10^{10}$ s and increased by a factor `dtkoef` $= 1.2$ when all stability criteria described in Section 3.10.3 are satisfied.

Solid-velocity (CFL) constraint. The solid-velocity constraint (Section 3.10.3) applies a CFL-like marker displacement criterion with a Courant number $C_{\mathrm{CFL}} =$ `dxymax` $= 0.1$.

Composition-change constraint. The chemical evolution constraint (Section 3.10.3) limits the change in water content to `dnfmax` $= 100\ \mathrm{mol\,m^{-3}}$ per timestep.

Porosity-change constraint. The porosity evolution constraint (Section 3.10.3) restricts the maximum porosity change to `dphimax` $= 0.01$ per timestep.

4.4 Resolution and model size

The model domain covers $40\,\mathrm{km} \times 25\,\mathrm{km}$ and is discretized on a fully staggered grid (Fig. 2) with 81×51 grid points. A marker-in-cell scheme is used (Section 3.4) with four markers per cell initially.

4.5 Boundary and Initial Conditions

4.5.1 Boundary Conditions

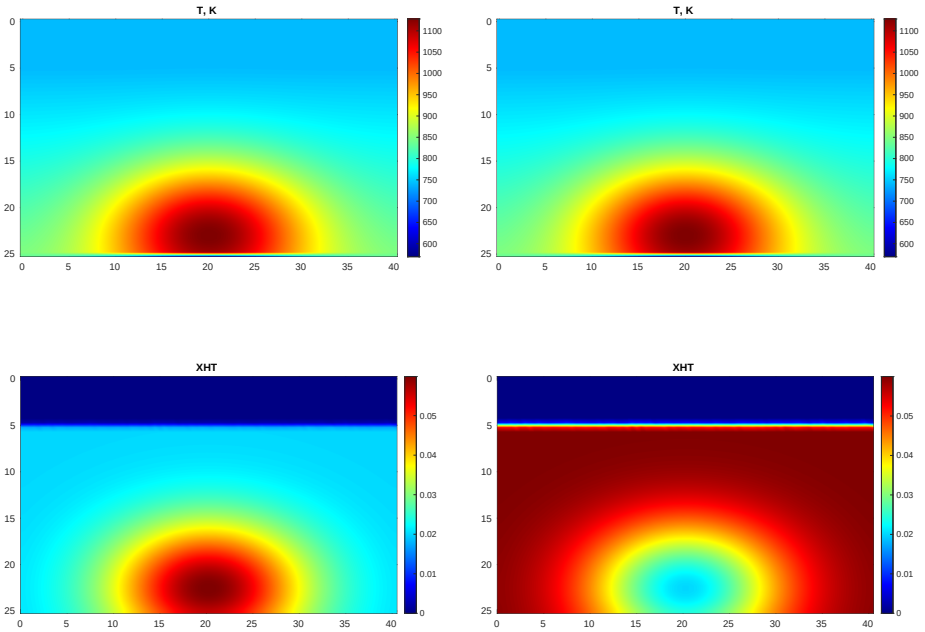
Thermal boundary conditions: Top to the surface: 740 K, Bottom: 840 K
Pressure boundary condition: 67.53×10^5 Pa $\rightarrow$ 95.6×10^5 Pa surface pressure
Mechanical boundary conditions: Free slip everywhere
Hydraulic boundary conditions: Free slip everywhere

4.5.2 Initial Conditions

We discuss two different intrusion scenarios with an intrusion modeled as a smooth elliptical Gaussian:

- **Wet magma intruding a dry crust:** Background hydration is 0.02 mol fraction, with a Gaussian shaped intrusion contributing an additional 0.04 mol fraction (Fig. 8a).
- **Dry magma intruding a wet crust:** Background hydration is 0.06 mol fraction, with a Gaussian shaped intrusion which is depleted by 0.04 mol fraction relative to the background (Fig. 8b).

Apart from the hydration configuration, the two cases do not differ. Surface temperature is 740 K, with a smooth gradient to 840 K at the bottom. An initial pressure guess with a smooth gradient is used to enable database lookup. The intrusion is centered horizontally in the middle at approximately 20 km depth, with a +300 K temperature anomaly. Initial viscosity is set to 1×10^{21} Pa·s in the crust and is updated with the flow-law parameters of plagioclase (An_{75}) according to Table 3. A viscosity value of 1×10^{17} Pa·s is used in the sticky air, layer. The Venera 14 crustal composition of Table 2 is used to compute the thermodynamic database. Initial solid hydration is determined via database lookup using the estimated pressure, initial temperature, and initial total hydration.



(a) Wet magma intruding a dry crust (b) Dry magma intruding a wet crust

Figure 8: Initial Temperature (T) and total hydration (XHT) of the two intrusion scenarios

5 Results and Discussion

5.1 Thermodynamic Database

5.1.1 Motivation

The HTMC numerical model requires material properties (density, enthalpy, heat capacity, thermal expansivity, compressibility) as continuous functions of pressure, temperature, and hydration. Computing these properties from Gibbs free energy minimization at every timestep would be computationally expensive. We therefore precompute a database of these properties and use trilinear interpolation to compute the values (Section 3.6).

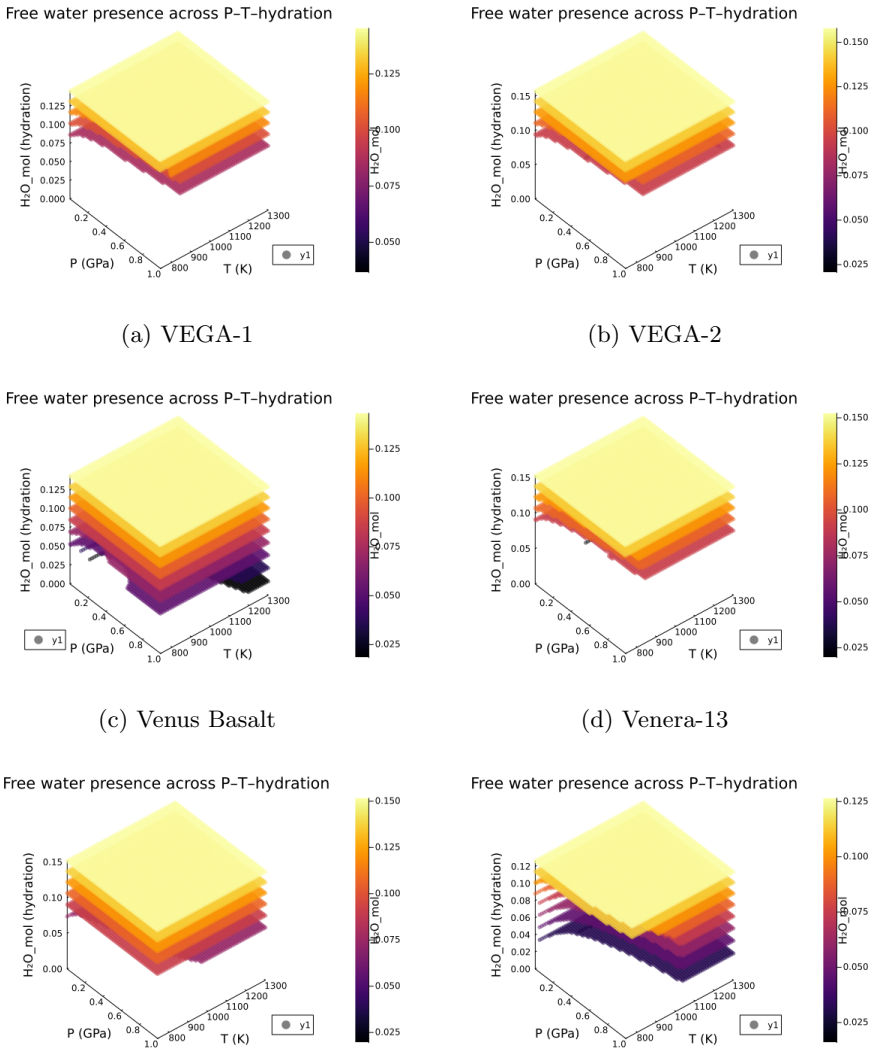
We focus on investigating water partitioning between structurally bound (solid-phase) and free (pore fluid) water, as this directly affects how the model behaves. Compositions with high water storage capacity in solids behave fundamentally differently from those where water remains as free fluid, since porous fluid transport via Darcy flow occurs much faster than solid advection.

We compute databases for all documented Venus compositions (Table 2), plus Earth's BSE composition [9] as reference. Database construction details are given in Section 3.2.

5.1.2 Results

Depending on the initial Venus crustal composition (Table 2), the computed database exhibits different thermodynamic properties and water storage capacities. Figure 9 shows the equilibrium free water content in moles as a function of pressure, temperature, and bulk water content for all six compositions. The databases of VEGA-1 (9a), VEGA-2 (9b), Venera-13 (9d), and Venera-14 (9e) show very similar presence of free water. For all four cases, free water starts to occur at around 0.08 ± 0.01 molar fraction of H_2O. At temperatures below 800 K they show that lower total hydration will not produce free water, but at higher hydration it will form free water. Overall, these four databases show quite similar results regarding free water content.

The Venus Basalt (9c) and BSE (9f) show different behavior than the other four cases. Even at low water content, free water is present, at least in high temperature regions. The Venus Basalt plot shows free water even at low temperatures when pressure is high.



(a) VEGA-1

(b) VEGA-2

(c) Venus Basalt

(d) Venera-13

(e) Venera-14

(f) BSE (Earth)

Figure 9: Equilibrium solid-phase water content $X_{\mathrm{H_2O}}^{\mathrm{solid}}$ as a function of depth, temperature, and bulk water content for six crustal compositions. Warmer colors indicate higher water storage capacity in solid phases. Compositional variations result in different mineralogies and water solubilities, which directly impact porosity evolution in the simulations.

5.1.3 Discussion

The thermodynamic database results show that, despite differences in bulk composition, most Venus landing-site based models exhibit similar water partitioning behavior. The VEGA-1, VEGA-2, Venera-13, and Venera-14 compositions all show comparable thresholds for the appearance of free water, with free fluid generally occurring only once bulk hydration exceeds approximately 0.08 ± 0.01 mol fraction. Below this threshold, water is predominantly stored in solid phases, even at higher temperatures.

The Venus Basalt composition stands out as an outlier. In contrast to the landing-site based models, it shows the presence of free water even at relatively low bulk hydration, particularly at high temperatures and pressures. This difference is likely related to how the Venus Basalt composition was constructed, as it is not tied to a specific landing site and therefore represents a more generalized basaltic composition rather than a directly measured surface sample. As such, its behavior cannot be directly compared to the Venera and VEGA compositions.

An interesting result is that, for the landing-site–based compositions, Venus does not exhibit large amounts of free water, despite the very high temperatures on Venus that would typically favor dehydration and fluid release. Instead, these compositions suggest a relatively high capacity for storing water in solid phases. Within the pressure–temperature range explored here, Venusian crustal material may therefore act as an efficient water reservoir.

It is important to emphasize that the hydration state of Venus and its crust is unknown. The hydration levels explored in these databases are theoretical ranges, chose,n to investigate the potential effects of different bulk water contents on phase equilibria and water partitioning. If Venus is extremely dry, with bulk hydration well below the thresholds shown here, little to no free water would be present and hydration-related processes would be negligible. The database cubes should therefore be viewed as a tool to explore how the system would behave if Venus had a given molar fraction of water.

5.2 HTMC evolution

5.2.1 Motivation

The interior structure and evolution of Venus remain largely unknown. Understanding Venus's interior would help to understand the planet and its past. While we possess limited compositional data from surface measurements (Table 1) used to construct our thermodynamic database (Section 3.2), as well as surface temperature and pressure measurements, much of our knowledge about Venus's interior must be inferred from orbital observations of surface features.

Surface features such as coronae. Large, quasi-circular volcanic and tectonic structures, provide evidence for magmatic intrusions into Venus's crust [3]. Thermomechanical modeling demonstrates that these coronae form through mantle plume impingement and magma-assisted crustal deformation, with intrusion dynamics spanning hundreds of thousands of years [3]. The mechanics

of magmatic intrusion are critically controlled by material properties of both the intruding magma and the crust [4].

The distribution of water between Venus's interior, crust, and atmosphere us unknown. In this thesis two scenarios are tested: **Wet magma intruding in a dry crust** and **Dry magma intruding wet crust**. To test whether these scenarios produce distinguishable signatures, we model these two cases identically except for the opposite hydration distributions (Section 4.5). Because direct measurements of crustal or magmatic hydration on Venus are not available, these hydration states should be regarded as hypothetical assumptions. Actual Venusian hydration levels may differ substantially.

5.2.2 Wet magma intruding in a dry crust evolution

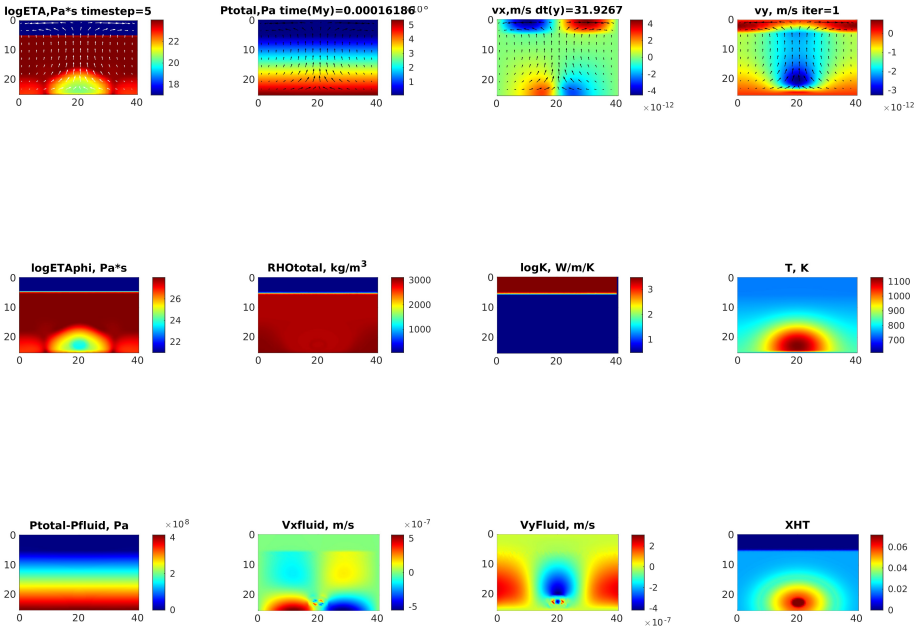
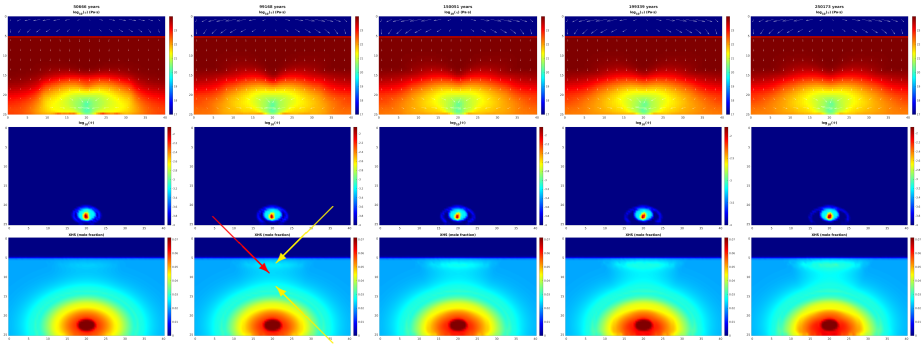
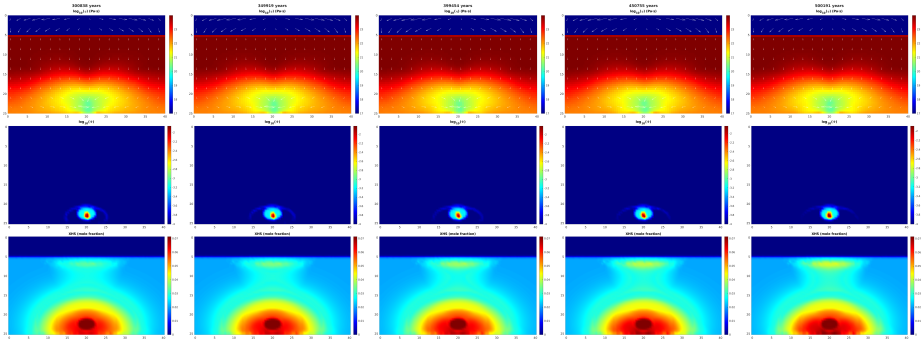


Figure 10: Wet magma intruding in a dry crust at the fifth timestep



(a) 50 kyr (b) 100 kyr (c) 150 kyr (d) 200 kyr (e) 250 kyr



(f) 300 kyr (g) 350 kyr (h) 400 kyr (i) 450 kyr (j) 500 kyr

Figure 11: Temporal evolution of viscosity (top), porosity (middle), and hydration (bottom) in the model at 50 kyr intervals.

Evolution of rheology, porosity and solid hydration Figure 11 shows the temporal evolution of wet magma intruding in a dry crust. What immediately stands out is that rheology and porosity (upper and middle plot) seem to change only slightly, if at all. Solid hydration, on the other hand, starts with the initial conditions as shown and described in Figure 8a and then begins to hydrate the solid at the top; in Figure 11b this starts to appear clearly. Over the next timesteps, solid hydration increases more and more in this area, reaching a maximum of ≈ 0.04 mol in Figure 11j, which is double the initial value. The two areas of higher hydration (indicated by the yellow arrows) begin to connect over the timesteps: in Figure 11b they appear to be weakly connected by hydration (indicated by the red arrow), and over time these areas merge together.

5.2.3 Dry magma intruding wet crust evolution

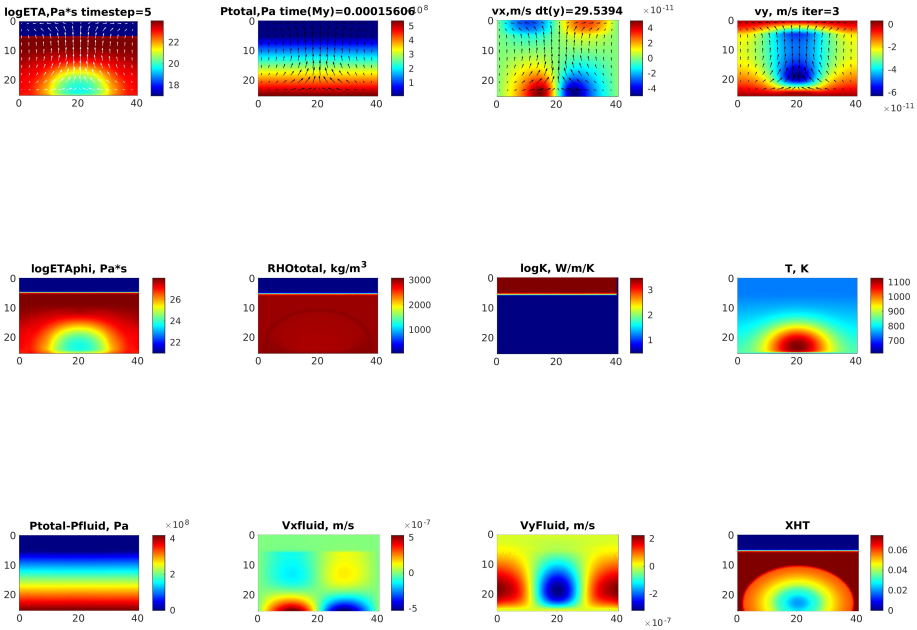
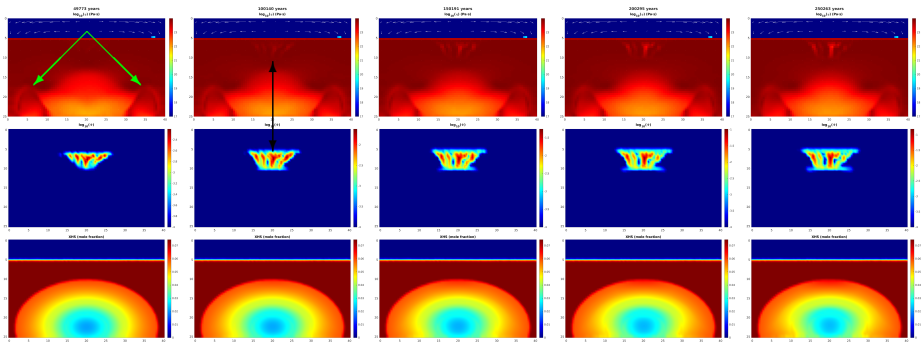
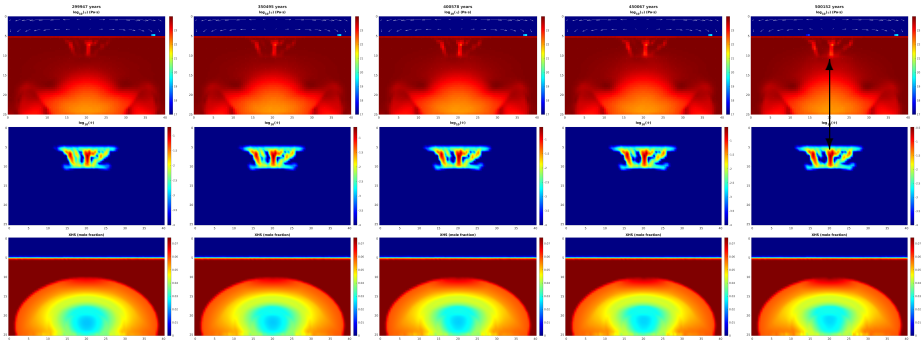


Figure 12: Fifth timestep evolution of background hydration 0.06 mol with 0.04 mol reduced gaussian plume at 20km depth



(a) 50 kyr (b) 100 kyr (c) 150 kyr (d) 200 kyr (e) 250 kyr



(f) 300 kyr (g) 350 kyr (h) 400 kyr (i) 450 kyr (j) 500 kyr

Figure 13: Temporal evolution of viscosity (top), porosity (middle), and hydration (bottom) in the model at 50 kyr intervals.

Evolution of rheology, porosity and solid hydration Figure 13 shows the evolution of dry magma intruding a wet crust. Comparing Figure 12, which shows the initial state of this configuration, with Figure 13a, representing the configuration at $\approx$ 50,000 years, reveals a large difference in rheology. It evolves from a smooth Gaussian at 20 km depth at timestep 5 to a smoothed-out Gaussian with two curved features to the left and right, indicated by the green arrows. From 50,000 years onward, the porosity and, with it, the rheology evolve slowly. The accumulation at the top, first visible in the rheology plot at 100,000 years (indicated by the black arrow, which illustrates how porosity affects rheology), continues to grow and becomes more pronounced over time, leading to the gradual formation of a surface rift. At 500,000 years, the black double arrows indicate the clearly developed structures. In contrast, outside the regions of accumulation, rheology and porosity, as well as solid hydration in the total frame, remain largely unchanged throughout the evolution.

5.2.4 Discussion

The two intrusion scenarios show different evolution despite being identical in all aspects except for the initial hydration distribution. In the case of wet magma intruding a dry crust, hydration primarily affects the solid phase, while rheology and porosity remain largely unchanged over the modeled timescale. In contrast, dry magma intruding a wet crust leads to localized porosity accumulation and associated rheological weakening, eventually producing surface-scale structures.

These results illustrate that the initial distribution of water between magma and crust can have a large influence on the mechanical and hydraulic evolution of the system. However, given the unknown hydration state of Venus and the simplifying assumptions of the model, the simulations should be interpreted as qualitative experiments that demonstrate the capabilities of the modeling framework rather than as predictions of Venusian crustal behavior.

All `.mat` files saved at every 50th timestep are available online [6].

5.3 Model Limitations

Since this is only a model with assumptions, there are several limitations where the model fails, is inaccurate, or produces unphysical or wrong solutions. The limitations arise from different factors such as discretization error, limited computing precision, assumptions, uncertain/limited data, etc.

Porosity cutoff The mechanical porosity ϕ has cutoff values of $\phi_{\min}$ and $\phi_{\max}$. These cutoff values are needed because there are terms proportional to $(1-\phi)^{-1}$ and ϕ^{-1} in the governing equations. Allowing porosity values close to 0 or 1 would lead to numerical issues: as $\phi \to 1$ or $\phi \to 0$, these terms become unbounded, resulting in divisions by very small numbers or by zero. Such behavior would lead to numerical instabilities and ill-conditioned linear systems. By evolving an unconstrained compositional porosity and using a bounded mechanical porosity in the governing equations, singular behavior is avoided. While these cutoffs ensure stability, they can lead to unphysical behavior because there will always be some porosity since $\phi_{\min} > 0$, which means that porosities $\phi < \phi_{\min}$ cannot be handled in the model. Furthermore, there will always be permeability because porosity can never reach 0.

Mass transfer rate Mass transfer rate of the hydration/dehydration reaction was not considered in this code. However, the framework exists with the stencils and DMP matrices, but they are initialized to 0 and never changed. So implementing mass transfer would not require much work.

Hydrous melting Although the previous code [19] took hydrous melting into account, it is not considered in the present model. The chemical formulation accounts only for the partitioning of H_2O between free pore fluid and structurally bound water in the solid. As a result, the model is not applicable to conditions

where partial melting and melt–fluid interactions play a significant role. Hydrous melting needs a melting model included in the database and tracing the evolution of the moving melt composition and thus evolution of the chemical composition of the system in addition to changes in total water content. This would also require extensive thermodynamic mapping in P-T-H_2O-SiO_2-Al_2O_3-etc. space.

5.3.1 Thermodynamic database

The thermodynamic database exhibits a handful of important limitations:

Venus composition Depending on the selected Venus composition (Table 2), results can be very different. Some compositions seem to produce more unstable databases and general code stability, whilst others are very stable. All of the compositions are modelled, inferred, measured locally, or a mix of those; these processes naturally carry errors with themselves.

Heat capacity Total heat capacity $c_{p,\mathrm{tot}}$ is computed directly from the database and not computed in the code, so volumetric effects are neglected.

Limits of the database In general, the ranges of pressure, temperature, and hydration are constrained by the limits of the precomputed database. Values outside these ranges are clipped to the corresponding minimum or maximum database values. As a result, the applicability of the model is restricted to the parameter space over which the database is defined. A larger parameter space will automatically lead to a larger database. In this project, the maximum hydration included in the database is limited to a molar fraction of 0.14. The resolution of the database in pressure, temperature, and hydration may also influence the accuracy of the model; however, the sensitivity to database resolution was not systematically tested. When input values fall outside the database range, the clipping procedure prevents the code from failing, but it is likely to introduce significant errors in the computed results. Users should therefore ensure that all model inputs remain within the validated bounds of the database.

5.3.2 Additional limitations

Several other limitations should be noted:

- **Instantaneous equilibrium assumption**: The model assumes instantaneous chemical equilibrium for water partitioning between solid and fluid phases, which may not be realistic for all conditions.
- **Grid resolution effects**: The spatial resolution (Nx=81, Ny=51) may not capture all relevant length scales of the physical processes.

- **Viscosity cutoffs**: The artificial bounds on viscosity ($\eta_{\min}$ and $\eta_{\max}$) can affect the dynamics in regions where the actual viscosity would fall outside these bounds.
- **2D limitation**: The model is two-dimensional and cannot capture three-dimensional effects such as out-of-plane flow or variations.
- **Boundary condition artifacts**: The choice of boundary conditions (free-slip, no-slip, insulating) can influence the solution, particularly near boundaries.

6 Conclusions

6.1 Summary

This thesis focused on developing a hydro-thermo-mechanical-chemical (HTMC) modeling framework to investigate potential crustal hydration processes on Venus. Because direct constraints on Venusian hydration, interior structure, and compositions are extremely limited, the goal of the project was less to obtain geologic conclusions and more to establish a working numerical setup that can test different scenarios.

A thermodynamic database approach was implemented to provide pressure-temperature-hydration dependent material properties through trilinear interpolation, avoiding expensive Gibbs minimization during runtime. This makes it possible to explore how different assumed hydration levels and different Venus crustal compositions affect water partitioning between solid-bound and free pore water, and how this partitioning feeds back on porosity, density, and rheology.

The database results show that the landing-site based compositions (Venera and Vega) behave relatively similarly in terms of free water presence, while the synthetic Venus Basalt composition behaves as an outlier, likely because it is constructed differently and is not landing-site specific.

Two intrusion scenarios (wet magma into dry crust and dry magma into wet crust) were tested to demonstrate the capability of the framework and to evaluate whether opposite hydration structures can produce distinguishable evolution patterns, but the results should be interpreted as exploratory because the absolute hydration state of Venus is unknown and several simplifying assumptions remain in the current implementation.

6.2 Future Work

Future work could either focus on systematically exploring the parameter space, for example by testing different geometries, initial conditions, resolutions, and comparing how the different thermodynamic databases affect the simulations. Alternatively, future studies could focus on improving specific model limitations or assumptions, such as the porosity cutoff or the instantaneous equilibrium assumption.

Another interesting direction would be a more detailed investigation of the thermodynamic databases themselves. The results show notable differences in free-water presence and suggest a relatively large water storage capacity for some Venus compositions. Further analysis of these databases could therefore help to better constrain plausible Venusian surface compositions and hydration states.

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